The Political Economy of Central-Bank Capture

Thresholds, Fiscal Dominance, Credibility, and Institutional Hysteresis

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Abstract

This paper develops a general dynamic theory of political capture of a central bank. Central-bank independence is modeled not as an immutable constitutional characteristic but as an endogenous state variable produced by strategic interaction among governments, monetary authorities, financial institutions, households, markets, legislatures, courts, and other domestic and international actors. The government may invest in capture through appointments, legal modification, budgetary pressure, dismissal threats, informational control, administrative reorganization, or other institutional mechanisms. The central bank simultaneously chooses monetary, supervisory, liquidity, reserve-management, and communication policies.

The model generates five principal results. First, sufficiently strong strategic complementarities create a Capture Threshold separating a low-capture basin from a high-capture basin. Second, fiscal stress can endogenously increase the political return to monetary capture, producing a Fiscal-Dominance Capture mechanism. Third, political capture can increase inflation and sovereign risk before observable monetary accommodation occurs through a Credibility-Destruction channel. Fourth, capture that damages the institutions protecting future independence generates Institutional Hysteresis: removal of the initiating political shock need not restore the original equilibrium. Fifth, multidimensional capture is governed locally by a Jacobian whose spectral radius provides a Systemic Capture Criterion.

The framework is extended using ideas from welfare economics, political science, international relations, conflict economics, control theory, statistical physics, chemistry, evolutionary biology, psychology, organizational behavior, philosophy, statistics, computer science, network science, data science, and machine learning. These cross-disciplinary connections are used as mathematical or conceptual analogies rather than as claims that political institutions literally constitute physical, biological, chemical, or psychiatric systems. An empirical architecture is proposed for estimating latent capture, detecting structural breaks, forecasting institutional transitions, and distinguishing ordinary democratic oversight from institutional subordination.

The paper ends with "The End"

Keywords: central-bank independence; political capture; fiscal dominance; monetary policy; institutional hysteresis; credibility; political economy; dynamic games; regime switching; systemic risk; machine learning.

1 Introduction

Modern monetary economics frequently treats central-bank independence as an institutional parameter. Yet independence is itself the outcome of political institutions, appointment mechanisms, legal constraints, fiscal conditions, professional norms, public expectations, and strategic interaction.

The central proposition of this paper is therefore

$$\boxed{\text{Central-bank independence is an endogenous state variable.}}$$

Once this proposition is accepted, the political economy of monetary policy changes fundamentally. The government does not merely choose fiscal policy while an independent monetary authority chooses interest rates. Political actors may also choose how much effort to devote to changing the institution that chooses monetary policy.

The relevant sequence becomes

$$\begin{array}{c}\text{political}\\\text{incentives}\end{array} \to \begin{array}{c}\text{capture}\\\text{effort}\end{array} \to \begin{array}{c}\text{institutional}\\\text{control}\end{array} \to \text{policy} \to \begin{array}{c}\text{economic}\\\text{outcomes}\end{array} \to \begin{array}{c}\text{future}\\\text{political}\\\text{incentives}\end{array}.$$

The framework builds on the classical literature concerning rules versus discretion, time inconsistency, political business cycles, central-bank independence, fiscal dominance, institutional economics, and political selection [14, 3, 19, 1, 20, 17].

The novelty lies in making the degree of institutional capture itself a dynamic strategic variable.

2 Conceptual Distinctions

Political accountability and political capture are not synonymous.

Definition 1 (Democratic oversight). *Democratic oversight consists of lawful mechanisms by which elected institutions specify mandates, require transparency, review performance, and appoint officials subject to predetermined constitutional or statutory rules.*

Definition 2 (Political influence). *Political influence exists when political actors attempt to affect monetary policy while the monetary authority retains meaningful capacity to reject those demands.*

Definition 3 (Political capture). *Political capture occurs when political actors acquire sufficient control over the central bank's objectives, personnel, information, instruments, or constraints that implemented policy systematically moves toward the political principal's preferred policy relative to the bank's statutory or delegated objective.*

Thus,

$$\text{oversight} \neq \text{influence} \neq \text{capture}.$$

This distinction is essential for both theory and empirical measurement.

3 Players, States, and Timing

Consider an infinite-horizon stochastic game with players

$$\mathcal{P} = \{G, B, F, H\},$$

where G denotes the government, B the central bank, F the financial sector, and H households and other private agents.

The aggregate state is

$$\mathbf{s}_t = (\pi_t, y_t, b_t, q_t, c_t, I_t, z_t),$$

where:

π_t inflation;

y_t output gap;

b_t government debt-to-output ratio;

q_t financial stress;

c_t aggregate political capture;

I_t institutional protection of central-bank independence;

z_t exogenous economic, political, geopolitical, or financial shocks.

The timing is

$$\mathbf{s}_t \to e_t \to \mathbf{a}_t^B \to (\mathbf{a}_t^F, \mathbf{a}_t^H) \to \mathbf{s}_{t+1},$$

where e_t denotes governmental capture effort.

4 The Independent Central Bank

Let the independent central bank minimize

$$L_t^B = \frac{1}{2}\left[(\pi_t - \pi^*)^2 + \lambda_y y_t^2 + \lambda_q q_t^2\right].$$

Its intertemporal problem is

$$V^B(\mathbf{s}_t) = \min_{\mathbf{a}_t^B} \mathbb{E}_t \sum_{j=0}^{\infty} \beta_B^j L_{t+j}^B.$$

The vector of policy instruments may include

$$\mathbf{a}_t^B = (i_t, m_t, r_t, \ell_t, x_t, d_t),$$

where i_t is the policy rate, m_t balance-sheet policy, r_t regulatory policy, ℓ_t lender-of-last-resort policy, x_t foreign-exchange intervention, and d_t communication or disclosure policy.

Let the bank's preferred policy be

$$\mathbf{a}_t^{B,*} = \arg\min_{\mathbf{a}_t^B} V^B(\mathbf{s}_t).$$

5 Government Preferences

Let governmental utility be

$$U_t^G = \theta_y y_t - \frac{\theta_\pi}{2}(\pi_t - \bar{\pi}_G)^2 - \frac{\theta_q}{2} q_t^2 - \theta_b \Phi(b_t, i_t) + \theta_E E_t - K(e_t),$$

where E_t denotes electoral or political benefits.

Capture effort is costly:

$$K(e_t) = \frac{\kappa_t}{2} e_t^2.$$

The parameter κ_t is the institutional price of capture.

Let

$$\kappa_t = \kappa(J_t, T_t, L_t, M_t, N_t),$$

where J_t is judicial protection, T_t transparency, L_t legislative constraint, M_t public and market monitoring, and N_t professional and constitutional norms.

We assume

$$\frac{\partial \kappa}{\partial J} > 0, \qquad \frac{\partial \kappa}{\partial T} > 0, \qquad \frac{\partial \kappa}{\partial L} > 0, \qquad \frac{\partial \kappa}{\partial M} > 0, \qquad \frac{\partial \kappa}{\partial N} > 0.$$

The government solves

$$V^G(\mathbf{s}_t) = \max_{e_t} \left\{ U_t^G + \beta_G \mathbb{E}_t V^G(\mathbf{s}_{t+1}) \right\}.$$

6 Implemented Policy Under Capture

Let $\mathbf{a}_t^{G,*}$ denote the government's preferred central-bank policy.

The implemented policy is

$$\tilde{\mathbf{a}}_t^B = (1 - c_t)\mathbf{a}_t^{B,*} + c_t \mathbf{a}_t^{G,*},$$

where

$$c_t \in [0, 1].$$

Therefore,

$$c_t = 0 \Rightarrow \widetilde{\mathbf{a}}_t^B = \mathbf{a}_t^{B,*},$$

and

$$c_t = 1 \Rightarrow \widetilde{\mathbf{a}}_t^B = \mathbf{a}_t^{G,*}.$$

This gives capture an operational interpretation.
Define the preferred-policy wedge

$$\Delta\mathbf{a}_t = \mathbf{a}_t^{G,*} - \mathbf{a}_t^{B,*}.$$

Then

$$\widetilde{\mathbf{a}}_t^B = \mathbf{a}_t^{B,*} + c_t \Delta\mathbf{a}_t.$$

Hence

$$\frac{\partial \widetilde{\mathbf{a}}_t^B}{\partial c_t} = \Delta\mathbf{a}_t.$$

Capture matters economically precisely when the political and technocratic objectives differ.

7 Endogenous Capture Dynamics

Let

$$c_{t+1} = (1-\rho)c_t + \rho\Lambda(\Xi_t),$$

where

$$\Lambda(x) = \frac{1}{1+\exp(-x)}$$

and

$$\Xi_t = \alpha + \eta e_t + \psi b_t + \omega P_t + \gamma A_t - \xi J_t - \tau T_t - \nu M_t.$$

Here P_t measures political pressure and A_t appointment control.
Because

$$\Lambda'(x) = \Lambda(x)[1-\Lambda(x)],$$

we obtain

$$\frac{\partial c_{t+1}}{\partial e_t} = \rho\eta\Lambda(\Xi_t)[1-\Lambda(\Xi_t)].$$

For $\eta > 0$,

$$\frac{\partial c_{t+1}}{\partial e_t} > 0.$$

Similarly,

$$\frac{\partial c_{t+1}}{\partial b_t} > 0$$

when $\psi > 0$, whereas

$$\frac{\partial c_{t+1}}{\partial J_t} < 0, \qquad \frac{\partial c_{t+1}}{\partial T_t} < 0, \qquad \frac{\partial c_{t+1}}{\partial M_t} < 0.$$

8 Optimal Political Investment in Capture

Define the continuation value of capture as

$$\Omega_t = \mathbb{E}_t \left[\frac{\partial V^G_{t+1}}{\partial c_{t+1}} \right].$$

The government's first-order condition is

$$-\kappa_t e_t + \beta_G \Omega_t \frac{\partial c_{t+1}}{\partial e_t} = 0.$$

Therefore,

$$\boxed{\kappa_t e_t = \beta_G \rho \eta \Lambda(\Xi_t)[1 - \Lambda(\Xi_t)]\Omega_t.}$$

The left-hand side is the marginal institutional cost of capture. The right-hand side is the discounted marginal political return to institutional control.

Proposition 1 (Institutional Resistance)**.** *Suppose the second-order condition for the government's capture problem holds. An increase in the marginal institutional cost parameter κ_t weakly reduces optimal capture effort.*

Proof. Let the first-order condition be

$$F(e, \kappa) = \kappa e - MB(e) = 0,$$

where

$$MB(e) = \beta_G \Omega \frac{\partial c'}{\partial e}.$$

The second-order condition for the government's maximization implies

$$F_e = \kappa - MB'(e) > 0.$$

By the implicit-function theorem,

$$\frac{\partial e^*}{\partial \kappa} = -\frac{F_\kappa}{F_e} = -\frac{e^*}{\kappa - MB'(e^*)}.$$

For $e^* > 0$,

$$\boxed{\frac{\partial e^*}{\partial \kappa} < 0.}$$

□

9 The Capture Threshold

Consider the reduced-form dynamics

$$c_{t+1} = F(c_t).$$

A stationary capture equilibrium c^* satisfies

$$c^* = F(c^*).$$

Local stability requires

$$|F'(c^*)| < 1.$$

Strategic complementarities can make F S-shaped. Suppose three fixed points exist:

$$c_L < c^\dagger < c_H.$$

Theorem 1 (Capture Threshold Theorem)**.** *Suppose*

$$|F'(c_L)| < 1,$$

$$F'(c^\dagger) > 1,$$

and

$$|F'(c_H)| < 1.$$

Then c_L and c_H are locally stable fixed points, while $c^\dagger$ is unstable. For initial conditions in their respective basins of attraction,

$$c_0 < c^\dagger \Rightarrow c_t \to c_L,$$

whereas

$$c_0 > c^\dagger \Rightarrow c_t \to c_H.$$

Proof. Define deviations from a fixed point by

$$\widehat{c}_t = c_t - c^*.$$

A first-order Taylor expansion gives

$$\widehat{c}_{t+1} = F'(c^*)\widehat{c}_t + o(\widehat{c}_t).$$

Therefore sufficiently small deviations decay geometrically whenever

$$|F'(c^*)| < 1.$$

They expand whenever

$$|F'(c^*)| > 1.$$

Applying these conditions to $c_L, c^\dagger, c_H$ establishes local stability of the outer equilibria and instability of the middle equilibrium. Continuity of the one-dimensional transition map establishes the corresponding local basins separated by the unstable threshold.

□

The theorem implies that institutional deterioration need not be linear. Political pressure that appears reversible below $c^\dagger$ can become self-reinforcing after the threshold is crossed.

10 Fiscal Dominance and Capture

Government debt evolves according to

$$b_{t+1} = \frac{1+i_t}{1+g_t} b_t + d_t - s_t,$$

where g_t is nominal output growth, d_t the primary deficit ratio, and s_t seigniorage or transfers from the monetary authority.

Then

$$\frac{\partial b_{t+1}}{\partial i_t} = \frac{b_t}{1+g_t} > 0.$$

Suppose capture lowers the contemporaneous policy rate preferred by the government:

$$\frac{\partial i_t}{\partial c_t} < 0.$$

Higher debt then increases the fiscal value of influencing monetary policy.

Theorem 2 (Fiscal-Dominance Capture Theorem)**.** *Suppose:*

1. *higher policy rates increase expected government financing costs;*
2. *greater capture moves policy toward the government's preferred interest rate;*
3. *the government's capture problem satisfies its second-order condition.*

Then, whenever the fiscal benefit dominates other effects,

$$\boxed{\frac{\partial e_t^*}{\partial b_t} > 0.}$$

Proof. Let

$$\Omega_t = \mathbb{E}_t \left[\frac{\partial V_{t+1}^G}{\partial c_{t+1}} \right].$$

The marginal fiscal value of capture contains the term

$$-\theta_b \frac{\partial \Phi}{\partial i} \frac{\partial i}{\partial c}.$$

Since

$$\frac{\partial \Phi}{\partial i} > 0$$

and

$$\frac{\partial i}{\partial c} < 0,$$

this component is positive.

Moreover, the sensitivity of debt-service costs to the interest rate rises with outstanding debt. Thus, under the stated dominance condition,

$$\frac{\partial \Omega_t}{\partial b_t} > 0.$$

Implicit differentiation of the government's first-order condition then gives

$$\frac{\partial e_t^*}{\partial b_t} > 0.$$

□

This result differs from conventional fiscal dominance. Fiscal stress does not merely constrain monetary policy; it alters the political incentive to change the institution controlling monetary policy.

11 Credibility Destruction

Suppose inflation expectations satisfy

$$\mathbb{E}_t \pi_{t+1} = \pi^* + \chi c_t + \zeta \mathbb{E}_t (c_{t+1} - c_t),$$

where

$$\chi > 0, \qquad \zeta > 0.$$

The first term represents the inflation target, the second the level effect of capture, and the third the anticipated institutional deterioration effect.

Let the Phillips curve be

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa_y y_t + u_t.$$

Proposition 2 (Credibility-Destruction Proposition)**.** *If private agents rationally associate political capture with a greater probability of future inflationary accommodation, an increase in capture can raise current inflation before contemporaneous monetary accommodation occurs.*

Proof. Holding the output gap and contemporaneous shock constant,

$$\frac{\partial \pi_t}{\partial c_t} = \beta \frac{\partial \mathbb{E}_t \pi_{t+1}}{\partial c_t}.$$

If

$$\frac{\partial \mathbb{E}_t \pi_{t+1}}{\partial c_t} > 0,$$

then

$$\boxed{\frac{\partial \pi_t}{\partial c_t} > 0.}$$

Thus the expectations channel alone is sufficient.

□

Capture therefore operates through two distinct mechanisms:

$$\boxed{\text{capture} \to \begin{cases} \text{implemented-policy channel,} \\ \text{expectations and credibility channel.} \end{cases}}$$

12 The Capture Paradox

Consider a stabilizing interest-rate rule

$$i_t^{\text{stab}} = r_t^* + \mathbb{E}_t \pi_{t+1} + \phi_\pi (\pi_t - \pi^*) + \phi_y y_t.$$

Differentiating with respect to capture gives

$$\frac{\partial i_t^{\text{stab}}}{\partial c_t} = \frac{\partial \mathbb{E}_t \pi_{t+1}}{\partial c_t} + \phi_\pi \frac{\partial \pi_t}{\partial c_t} + \phi_y \frac{\partial y_t}{\partial c_t}.$$

Proposition 3 (Capture Paradox)**.** *If the inflation-expectations and inflation effects of capture dominate its output-gap effect, then*

$$\frac{\partial i_t^{\text{stab}}}{\partial c_t} > 0.$$

Political intervention intended to obtain lower short-run interest rates can therefore increase the interest rate ultimately required for stabilization.

The mechanism is

$$\boxed{c \uparrow \to i_{\text{political}} \downarrow \to \text{credibility} \downarrow \to \mathbb{E}\pi \uparrow \to \pi \uparrow \to i_{\text{stabilizing}} \uparrow.}$$

13 Sovereign Finance and the Capture–Debt Doom Loop

Let the sovereign yield be

$$R_t = r_t^* + \mathbb{E}_t \pi_{t+1} + \mu_t,$$

where the risk premium is

$$\mu_t = \mu_0 + \mu_b b_t + \mu_c c_t + \mu_q q_t.$$

Then

$$\frac{\partial R_t}{\partial c_t} = \frac{\partial \mathbb{E}_t \pi_{t+1}}{\partial c_t} + \mu_c.$$

For positive credibility and institutional-risk premia,

$$\frac{\partial R_t}{\partial c_t} > 0.$$

Consequently,

$$b_t \uparrow \Rightarrow e_t^* \uparrow \Rightarrow c_{t+1} \uparrow \Rightarrow \mathbb{E}_t \pi_{t+1} \uparrow \Rightarrow R_t \uparrow \Rightarrow b_{t+1} \uparrow .$$

This produces the Capture–Debt Doom Loop:

$$\boxed{b \to e \to c \to \text{credibility loss} \to R \to b.}$$

14 Institutional Hysteresis

Capture can alter the institutions that constrain future capture.

Let

$$I_{t+1} = (1 - \delta_I) I_t - \delta_c c_t + \nu X_t,$$

where X_t represents institutional reinforcement.

Therefore,

$$\frac{\partial I_{t+1}}{\partial c_t} = -\delta_c < 0.$$

Suppose

$$c_{t+1} = F(c_t, I_t)$$

with

$$F_I < 0.$$

Then

$$c_t \uparrow \Rightarrow I_{t+1} \downarrow \Rightarrow c_{t+2} \uparrow .$$

Theorem 3 (Institutional Hysteresis Theorem)**.** *Suppose capture reduces future institutional protection, lower institutional protection increases subsequent capture, and the resulting dynamics contain multiple stable equilibria. Then there exist temporary political shocks that permanently move the economy from the basin of a low-capture equilibrium to the basin of a high-capture equilibrium.*

Proof. Let the joint system be

$$\begin{pmatrix} c_{t+1} \\ I_{t+1} \end{pmatrix} = \mathcal{F}\left(\begin{pmatrix} c_t \\ I_t \end{pmatrix}, \epsilon_t \right).$$

Assume two locally stable stationary equilibria,

$$(c_L, I_H)$$

and

$$(c_H, I_L),$$

with $c_L < c_H$ and $I_H > I_L$.

Let $\mathcal{B}_L$ and $\mathcal{B}_H$ denote their basins of attraction. Because the basin boundary is nonempty, there exists a finite shock sequence

$$\{\epsilon_t\}_{t=0}^{T}$$

that moves the state from $\mathcal{B}_L$ into $\mathcal{B}_H$.

Set

$$\epsilon_t = 0 \qquad \forall t > T.$$

Since the post-shock state lies in $\mathcal{B}_H$,

$$\lim_{t\to\infty}(c_t, I_t) = (c_H, I_L),$$

rather than (c_L, I_H). Thus removal of the initiating shock does not restore the original equilibrium. $\square$

15 Multidimensional Political Capture

Scalar capture can conceal important institutional differences. Define

$$\mathbf{c}_t = \begin{pmatrix} c_t^M \\ c_t^S \\ c_t^L \\ c_t^A \\ c_t^D \\ c_t^X \end{pmatrix},$$

where the components represent capture of monetary policy, supervision, lender-of-last-resort policy, appointments, disclosure and communication, and foreign-exchange or reserve policy.

Aggregate capture may be summarized by

$$c_t = \boldsymbol{w}'\mathbf{c}_t,$$

where

$$w_j \geq 0, \qquad \sum_j w_j = 1.$$

The structural dynamics are

$$\mathbf{c}_{t+1} = \mathbf{F}(\mathbf{c}_t, \mathbf{e}_t, \mathbf{I}_t, \mathbf{z}_t).$$

Linearizing around an equilibrium $\mathbf{c}^*$ gives

$$\widehat{\mathbf{c}}_{t+1} = J_C\widehat{\mathbf{c}}_t + J_e\widehat{\mathbf{e}}_t + J_I\widehat{\mathbf{I}}_t + J_z\widehat{\mathbf{z}}_t,$$

where

$$J_C = \left.\frac{\partial \mathbf{F}}{\partial \mathbf{c}'}\right|_{\mathbf{c}^*}.$$

Theorem 4 (Systemic Capture Criterion)**.** *For the autonomous local system*

$$\widehat{\mathbf{c}}_{t+1} = J_C\widehat{\mathbf{c}}_t,$$

the equilibrium is locally asymptotically stable if

$$\boxed{\rho(J_C) < 1,}$$

where $\rho(J_C)$ is the spectral radius. If

$$\rho(J_C) > 1,$$

there exists at least one locally expanding institutional mode.

Proof. The solution is

$$\widehat{\mathbf{c}}_t = J_C^t\widehat{\mathbf{c}}_0.$$

A finite-dimensional discrete linear system converges to zero for every initial condition if and only if every eigenvalue of J_C lies strictly inside the unit circle. Equivalently,

$$\rho(J_C) < 1.$$

If $\rho(J_C) > 1$, an eigenvalue λ exists with $|\lambda| > 1$. A nonzero projection of the initial disturbance onto the corresponding generalized eigenspace grows rather than vanishes.

$\square$

16 Network Representation of Institutional Capture

Let institutions form a directed network

$$\mathcal{G} = (V, E).$$

Nodes may include the executive, legislature, courts, treasury, central bank, commercial banks, regulators, statistical agencies, media, and international institutions.

Let A be the weighted adjacency matrix.

A linearized propagation model is

$$\mathbf{x}_{t+1} = \alpha A \mathbf{x}_t + \boldsymbol{\epsilon}_t.$$

Institutional shocks decay if

$$\alpha\rho(A) < 1.$$

They can amplify when

$$\alpha\rho(A) > 1.$$

This gives a network interpretation of systemic capture: institutional fragility depends not only on the weakness of individual nodes but also on the topology of dependencies among them.

17 Control-Theoretic Interpretation

The central bank may be viewed mathematically as a feedback controller, while the macroeconomy is the controlled system.

Consider

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + B\mathbf{a}_t + D\mathbf{z}_t,$$

with policy rule

$$\mathbf{a}_t = -K_B \mathbf{x}_t$$

under independence.

Under capture,

$$K(c_t) = (1 - c_t)K_B + c_t K_G.$$

Hence

$$\mathbf{x}_{t+1} = [A - BK(c_t)]\mathbf{x}_t + D\mathbf{z}_t.$$

Stability requires

$$\rho[A - BK(c_t)] < 1.$$

This yields a macroeconomic stability threshold

$$c < c_{\text{stab}}$$

whenever sufficiently large capture moves an eigenvalue outside the unit circle.

Political capture can therefore be interpreted as endogenous modification of the economy's feedback controller.

18 Statistical-Physics Analogy

The capture-threshold model has a useful mathematical analogy to phase transitions.

Define an institutional potential function

$$V(c) = \frac{a}{4}c^4 - \frac{b}{3}c^3 + \frac{d}{2}c^2 - hc.$$

Let capture follow

$$\dot{c} = -\frac{\partial V}{\partial c} + \sigma\epsilon_t.$$

Stationary deterministic states satisfy

$$\frac{\partial V}{\partial c} = 0.$$

When V possesses two local minima separated by a local maximum, the system has two metastable institutional regimes.

The political-pressure parameter h tilts the potential landscape.

The analogy is mathematical, not ontological: political institutions are not physical particles. The usefulness of the analogy lies in metastability, threshold crossing, path dependence, and critical transitions.

19 Chemical-Reaction Analogy

A second useful analogy comes from autocatalytic chemical reactions.

Let C denote captured institutional capacity and I independent capacity. Consider the stylized transformation

$$I + C \longrightarrow 2C.$$

If capture facilitates additional capture, a reduced-form kinetic equation is

$$\dot{c} = k_1 c(1-c) - k_2 c.$$

The first term represents self-reinforcing conversion and the second institutional restoration.

More generally,

$$\dot{c} = k_0 + k_1 c^n(1-c) - k_2 c,$$

where $n > 1$ permits nonlinear feedback and multiple stationary states.

Again, this is a mathematical analogy. Political institutions do not obey chemical laws. The analogy isolates the logic of autocatalysis.

20 Evolutionary and Biological Analogy

Institutional norms may be represented as competing behavioral strategies.

Let p_t denote the proportion of strategically relevant actors supporting central-bank independence. Let the payoff difference between independence and political subordination be

$$\Delta U(p,c) = U_I(p,c) - U_C(p,c).$$

A replicator equation is

$$\dot{p} = p(1-p)\Delta U(p,c).$$

Capture may reduce the private payoff to defending independence:

$$\frac{\partial \Delta U}{\partial c} < 0.$$

Thus,

$$c \uparrow \Rightarrow \Delta U \downarrow \Rightarrow \dot{p} \downarrow .$$

This formalizes erosion of professional norms as an evolutionary coordination problem.

The biological language is analogical: institutional actors are strategic agents rather than genes or organisms.

21 Psychology and Organizational Behavior

Political capture also has microfoundations in individual decision-making.

Let a central-bank official choose resistance $r_i \in [0, 1]$.

Suppose utility is

$$U_i = \alpha_i r_i - \frac{\beta_i}{2} r_i^2 - P_i(c) r_i + R_i r_i,$$

where α_i represents intrinsic commitment to institutional independence, $P_i(c)$ expected personal or professional punishment, and R_i reputational rewards for resistance.

The first-order condition is

$$\alpha_i - \beta_i r_i - P_i(c) + R_i = 0.$$

Hence

$$r_i^* = \frac{\alpha_i + R_i - P_i(c)}{\beta_i},$$

subject to the bounds $0 \leq r_i^* \leq 1$.

If

$$P_i'(c) > 0,$$

then

$$\frac{\partial r_i^*}{\partial c} < 0.$$

Capture can consequently become self-reinforcing because increasing institutional control changes the incentives of officials who would otherwise resist it.

Psychological mechanisms that may contribute include conformity, authority effects, preference falsification, motivated reasoning, group polarization, career concerns, and pluralistic ignorance [2, 16, 13].

These mechanisms describe possible behavioral channels. They do not justify psychiatric diagnosis of political actors. Psychiatric categories concern clinical assessment of individuals and should not be inferred merely from institutional or political behavior.

22 Philosophy of Delegation and Institutional Authority

The problem also contains a normative philosophical tension.

A democratic state delegates authority to a technocratic institution because commitment can improve intertemporal welfare. Yet delegation cannot imply that the institution becomes sovereign over the democratic state.

Let

$$D = \text{democratic responsiveness}$$

and

$$K = \text{credible commitment}.$$

A stylized constitutional objective is

$$W^{\text{const}} = W(D, K).$$

Complete political discretion may produce high contemporaneous responsiveness but weak commitment. Complete technocratic insulation may produce commitment but insufficient democratic accountability.

The institutional design problem is therefore not

$$\max K,$$

but rather

$$\max_{\mathcal{I}} W[D(\mathcal{I}), K(\mathcal{I})],$$

where $\mathcal{I}$ is the institutional architecture.

Thus central-bank independence is properly understood as delegated, rule-governed authority rather than absence of democratic legitimacy.

23 Political Science

Political capture depends on veto points, party structure, executive power, constitutional rigidity, judicial independence, appointment rules, legislative oversight, and bureaucratic autonomy.

Let political concentration be P_t and institutional veto capacity be V_t.

A simple capture technology is

$$c_{t+1} = \Lambda(\alpha + \beta_P P_t - \beta_V V_t + \eta e_t).$$

Then

$$\frac{\partial c_{t+1}}{\partial P_t} > 0,$$

while

$$\frac{\partial c_{t+1}}{\partial V_t} < 0.$$

Political fragmentation may therefore reduce the feasibility of capture even when individual political actors would prefer greater monetary control.

24 International Relations and Geopolitics

Central-bank independence may also interact with international constraints.

Let foreign reserve pressure be f_t, exchange-rate pressure x_t, sanctions pressure s_t, and geopolitical conflict intensity g_t.

The external constraint can be represented by

$$\mathcal{E}_t = \omega_f f_t + \omega_x x_t + \omega_s s_t + \omega_g g_t.$$

Capture incentives may satisfy

$$e_t^* = e^*(b_t, \mathcal{E}_t, I_t).$$

The sign of

$$\frac{\partial e_t^*}{\partial \mathcal{E}_t}$$

is theoretically ambiguous.

External pressure can increase capture incentives if the government seeks direct monetary financing or capital allocation. Conversely, external discipline can increase the cost of capture if exchange-rate depreciation, capital flight, treaty obligations, or market access make monetary credibility especially valuable.

This yields a conditional geopolitical hypothesis:

$$\text{external pressure} \rightarrow \begin{cases} \text{greater capture,} & \text{if fiscal-security demands dominate,} \\ \text{greater independence,} & \text{if credibility constraints dominate.} \end{cases}$$

25 Warfare Economics and National Emergency

War and severe national emergencies alter the government's intertemporal budget constraint.

Let military expenditure be G_t^W. Then

$$B_{t+1} = (1 + i_t)B_t + G_t^C + G_t^W - T_t - S_t,$$

where G_t^C is civilian expenditure and S_t monetary financing.

Define the emergency state

$$w_t \in \{0, 1\}.$$

The political value of monetary control may be

$$\Omega_t = \Omega_0 + \Omega_b b_t + \Omega_w w_t.$$

If

$$\Omega_w > 0,$$

war increases the incentive to coordinate or subordinate monetary policy to fiscal-security objectives.

However, wartime coordination and political capture remain conceptually distinct. Temporary emergency coordination governed by explicit law, transparent objectives, and credible sunset provisions need not constitute institutional capture.

A useful distinction is therefore

$$\text{temporary policy coordination} \neq \text{permanent institutional subordination.}$$

26 Welfare Economics

Let representative-household welfare be

$$W_t^H = u(C_t) - v(N_t) - \frac{\omega_\pi}{2}(\pi_t - \pi^*)^2 - \frac{\omega_q}{2}q_t^2.$$

Social welfare is

$$\mathcal{W} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t W_t^H.$$

Capture may generate short-run gains if it offsets excessively restrictive policy or resolves coordination failures. It may also generate long-run costs through inflation, financial repression, credit misallocation, reduced credibility, and greater tail risk.

Define the welfare effect of marginal capture as

$$\frac{d\mathcal{W}}{dc} = \underbrace{\frac{\partial \mathcal{W}}{\partial y}\frac{\partial y}{\partial c}}_{\text{output channel}} + \underbrace{\frac{\partial \mathcal{W}}{\partial \pi}\frac{\partial \pi}{\partial c}}_{\text{inflation channel}} + \underbrace{\frac{\partial \mathcal{W}}{\partial q}\frac{\partial q}{\partial c}}_{\text{financial-stability channel}} + \underbrace{\frac{\partial \mathcal{W}}{\partial I}\frac{\partial I}{\partial c}}_{\text{institutional channel}}.$$

Consequently, the model does not impose the proposition that every marginal increase in political influence necessarily lowers welfare. The welfare question depends on the initial institutional equilibrium and on whether the intervention corrects or creates distortions.

27 Financial Repression and Credit Allocation

Capture need not operate through inflation alone.

Suppose regulated institutions hold government debt share h_t:

$$h_t = h_0 + \gamma_c c_t + \gamma_r r_t.$$

The sovereign financing cost may decline mechanically in the short run:

$$\frac{\partial R_t^{\text{short}}}{\partial h_t} < 0.$$

But forced portfolio allocation can distort capital allocation.

Let productive private investment be K_t^P. Then

$$K_t^P = S_t - B_t^H,$$

where B_t^H is captive sovereign-debt absorption.

If

$$\frac{\partial B_t^H}{\partial c_t} > 0,$$

then

$$\frac{\partial K_t^P}{\partial c_t} < 0.$$

Political capture may therefore substitute hidden financial repression for explicit monetary accommodation.

28 Information Capture

Suppose the public observes signal

$$s_t = \theta_t + \epsilon_t,$$

where θ_t is the true economic state.

Under information capture, the published signal becomes

$$\widetilde{s}_t = \theta_t + \delta(c_t) + \epsilon_t.$$

If

$$\delta'(c_t) \neq 0,$$

the information structure itself becomes endogenous.

Bayesian agents then form

$$\mathbb{E}[\theta_t \mid \widetilde{s}_t, c_t].$$

The central bank's credibility problem becomes more severe because agents must infer both the underlying economic state and the reliability of official communication.

29 A Latent-Variable Statistical Model

Capture is only partially observable. Let the latent capture state satisfy

$$c_t = \phi c_{t-1} + \boldsymbol{\gamma}' \mathbf{z}_t + u_t.$$

Observed indicators satisfy

$$\mathbf{y}_t = \boldsymbol{\Lambda} c_t + \boldsymbol{\Gamma} \mathbf{x}_t + \boldsymbol{\epsilon}_t.$$

Possible indicators include:

- unexpected governor or board-member removals;
- legal changes affecting tenure or appointment powers;
- deviations from estimated monetary-policy reaction functions;
- central-bank financing of government;

- directed credit or regulatory forbearance;
- unusual changes in communication;
- central-bank balance-sheet exposure to the sovereign;
- market-based inflation expectations;
- sovereign spreads;
- exchange-rate and capital-flow responses.

A state-space model can estimate

$$\widehat{c}_t = \mathbb{E}[c_t \mid \mathbf{y}_{1:t}].$$

30 Bayesian Filtering

Let the transition density be

$$p(c_t \mid c_{t-1})$$

and observation density

$$p(\mathbf{y}_t \mid c_t).$$

The prediction step is

$$p(c_t \mid \mathbf{y}_{1:t-1}) = \int p(c_t \mid c_{t-1}) p(c_{t-1} \mid \mathbf{y}_{1:t-1}) dc_{t-1}.$$

The updating step is

$$p(c_t \mid \mathbf{y}_{1:t}) \propto p(\mathbf{y}_t \mid c_t) p(c_t \mid \mathbf{y}_{1:t-1}).$$

This framework allows uncertainty about institutional capture to be represented explicitly rather than replacing the latent state by a deterministic index.

31 Regime-Switching Model

Define a latent institutional regime

$$S_t \in \{L, T, H\},$$

where L is low capture, T a transitional regime, and H high capture.
Let

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij,t}.$$

Transition probabilities can depend on fiscal and political variables:

$$p_{ij,t} = \frac{\exp(\alpha_{ij} + \boldsymbol{\beta}'_{ij}\mathbf{x}_t)}{\sum_k \exp(\alpha_{ik} + \boldsymbol{\beta}'_{ik}\mathbf{x}_t)}.$$

This allows direct empirical testing of the Capture Threshold hypothesis.

32 Structural-Break Detection

Suppose the monetary-policy reaction function is

$$i_t = \alpha + \phi_\pi \pi_t + \phi_y y_t + \phi_q q_t + \epsilon_t.$$

Capture may produce a parameter break:

$$\boldsymbol{\phi}_t = \begin{cases} \boldsymbol{\phi}^I, & t < \tau, \\ \boldsymbol{\phi}^C, & t \geq \tau. \end{cases}$$

A candidate break date can be estimated by

$$\widehat{\tau} = \arg\min_{\tau} \left[SSR_1(\tau) + SSR_2(\tau) \right].$$

The empirical challenge is identification: policy-rule changes may reflect legitimate mandate changes, model misspecification, crises, or new information rather than capture. Institutional evidence must therefore complement statistical breaks.

33 Causal Identification

A causal estimate of capture requires separating political intervention from macroeconomic conditions that independently affect monetary policy.

A panel specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta C_{it} + \boldsymbol{\gamma}' X_{it} + \epsilon_{it},$$

where Y_{it} may represent inflation expectations, sovereign spreads, inflation, reserve losses, or output volatility.

The coefficient β is causal only under appropriate identification conditions.

Potential strategies include:

1. event studies around plausibly exogenous institutional interventions;
2. difference-in-differences designs around legal reforms;
3. close-election or constitutional-threshold designs where justified;
4. instrumental variables based on predetermined appointment timing;
5. synthetic controls for major institutional regime changes;
6. local projections estimating dynamic responses.

A local projection can be written

$$Y_{i,t+h} - Y_{i,t-1} = \alpha_i^h + \lambda_t^h + \beta_h \Delta C_{it} + \boldsymbol{\gamma}_h' X_{it} + \epsilon_{i,t+h}.$$

The sequence

$$\{\beta_h\}_{h=0}^H$$

estimates the dynamic response to an institutional capture shock.

34 Machine Learning

Machine learning can supplement rather than replace structural analysis.

Let the feature vector be

$$X_t = (b_t, \pi_t, i_t, q_t, \text{appointments}, \text{legal changes}, \text{communications}, \text{market variables}, \text{political variables}).$$

Define a forecast target

$$Y_{t+h} = \mathbf{1}\{c_{t+h} > c^\dagger\}.$$

A classifier estimates

$$\widehat{p}_{t,h} = \mathbb{P}(c_{t+h} > c^\dagger \mid X_t).$$

Possible models include logistic regression, random forests, gradient boosting, neural networks, recurrent architectures, transformers, and state-space neural models.

For interpretability, a logistic benchmark is

$$\log\left(\frac{p_t}{1-p_t}\right) = \beta_0 + \boldsymbol{\beta}' X_t.$$

Machine-learning forecasts should be evaluated using strictly out-of-sample data.

35 Text Analysis and Natural-Language Processing

Central-bank communications, legislation, political speeches, and official statements contain information about institutional pressure.

Let document d_t be mapped into embedding

$$\mathbf{h}_t = f_\theta(d_t).$$

A capture-pressure score is

$$\widehat{P}_t = g_\omega(\mathbf{h}_t).$$

The score can enter the state equation:

$$c_{t+1} = F(c_t, e_t, \widehat{P}_t, I_t).$$

Supervised training requires carefully coded examples distinguishing criticism, legitimate oversight, policy disagreement, threats, personnel intervention, and institutional subordination.

36 Graph Machine Learning

If capture spreads across an institutional network, a graph representation is natural.

Let

$$H^{(0)} = X$$

contain node features.

A generic message-passing layer is

$$H^{(\ell+1)} = \sigma\left(\widetilde{A} H^{(\ell)} W^{(\ell)}\right),$$

where $\widetilde{A}$ is a normalized institutional adjacency matrix.

The final representation can estimate

$$\mathbb{P}(\text{systemic capture} \mid \mathcal{G}, X).$$

The model can potentially identify institutions that act as transmission channels or firebreaks.

37 Reinforcement-Learning Interpretation

The political problem can be represented as a Markov decision process.

The state is

$$s_t = (b_t, c_t, I_t, \pi_t, y_t, q_t, \ldots).$$

The political action is

$$a_t = e_t.$$

The reward is

$$r_t = U_t^G.$$

The action-value function satisfies

$$Q^\pi(s_t, e_t) = \mathbb{E}\left[r_t + \beta_G Q^\pi(s_{t+1}, e_{t+1})\right].$$

This interpretation is useful computationally for solving nonlinear versions of the model. It should not be interpreted as prescribing how governments ought to capture institutions.

38 Algorithm for Solving the Structural Model

A numerical solution can proceed by value-function iteration.

Algorithm 1: Dynamic Political-Capture Equilibrium

Input: Structural parameters Θ, state grid $\mathcal{S}$, capture-effort grid $\mathcal{E}$, tolerance ε.

Initialize: $V_0^G(s) = 0$ for every $s \in \mathcal{S}$.

Repeat:

1. For every state s:

   (a) Solve the independent central bank's policy problem and obtain $\mathbf{a}^{B,*}(s)$.

   (b) For each $e \in \mathcal{E}$, compute the next-period capture state

   $$c' = (1 - \rho)c + \rho\Lambda(\Xi).$$

   (c) Compute implemented monetary policy.

   (d) Solve private-sector equilibrium conditions.

   (e) Compute the transition distribution $p(s' \mid s, e)$.

   (f) Evaluate

   $$Q(s, e) = U^G(s, e) + \beta_G \sum_{s'} p(s' \mid s, e) V_n^G(s').$$

2. Set

   $$V_{n+1}^G(s) = \max_{e \in \mathcal{E}} Q(s, e).$$

3. Store

   $$e^*(s) = \arg\max_{e \in \mathcal{E}} Q(s, e).$$

Until

$$\max_s |V_{n+1}^G(s) - V_n^G(s)| < \varepsilon.$$

Output: $V^G(s)$, $e^*(s)$, $c'(s)$, policy functions, transition probabilities, and stationary distributions.

39 Algorithm for Empirical Early Warning

Algorithm 2: Central-Bank Capture Early-Warning System

Input: Macroeconomic, financial, institutional, legal, textual, and political data.

1. Construct a country-time panel.
2. Standardize observable indicators using only training-period information.
3. Estimate latent capture $\widehat{c}_{it}$ using a state-space or factor model.
4. Detect structural breaks in reaction functions and institutional variables.
5. Estimate the threshold $\widehat{c}^{\dagger}$.
6. Estimate transition probability

$$\widehat{p}_{i,t,h} = \mathbb{P}(c_{i,t+h} > \widehat{c}^{\dagger} \mid X_{it}).$$

7. Evaluate calibration, precision, recall, ROC-AUC, PR-AUC, Brier score, and false-alarm rates.
8. Perform rolling or expanding-window out-of-sample validation.
9. Stress-test predictions against alternative definitions of capture.
10. Report predictive uncertainty and causal uncertainty separately.

Output: A probabilistic institutional-risk indicator rather than a deterministic political classification.

40 Model Comparison

Table 1: Alternative Models of Central-Bank Political Influence

Model	Political variable	Main mechanism	Distinctive implication
Time inconsistency	Government preferences	Inflation surprise	Commitment improves equilibrium
Political business cycle	Election timing	Pre-election stimulus	Policy varies around elections
Fiscal dominance	Public debt	Fiscal constraint on monetary policy	Debt restricts monetary autonomy
Static capture	Political control	Policy distortion	Captured policy differs from delegated policy
Dynamic capture model	Endogenous c_t and e_t	Investment in institutional control	Thresholds, hysteresis, multiple equilibria

41 Multidisciplinary Correspondence

Table 2: Cross-Disciplinary Interpretation of the Capture Framework

Field	Mathematical concept	Central-bank-capture interpretation
Economics	Dynamic optimization	Government chooses costly capture effort
Finance	Risk premium	Institutional deterioration affects sovereign yields
Political science	Veto points	Institutional constraints raise capture costs
International relations	External constraint	Geopolitical shocks alter monetary-control incentives
Welfare economics	Social loss	Capture has output, inflation, stability, and institutional effects
Control theory	Feedback stability	Capture modifies the macroeconomic controller
Physics	Metastability	Low- and high-capture equilibria separated by thresholds
Chemistry	Autocatalysis	Existing capture facilitates additional capture
Evolutionary biology	Replicator dynamics	Professional norms survive or erode endogenously
Psychology	Conformity and incentives	Individual resistance falls when expected punishment rises
Philosophy	Delegation and legitimacy	Commitment must coexist with democratic accountability
Statistics	Latent-state inference	Capture is inferred from noisy indicators
Computer science	Dynamic programming	Equilibrium solved recursively
Machine learning	Classification	Estimate probability of threshold crossing
Network science	Spectral radius	Capture propagates through institutional dependencies

42 Integrated Dynamic System

The principal mechanisms can be collected into

$$\begin{aligned} c_{t+1} &= F(c_t, e_t, I_t, b_t, P_t, z_t), \\ I_{t+1} &= G(I_t, c_t, X_t), \\ b_{t+1} &= H(b_t, i_t, d_t, s_t, g_t), \\ \mathbb{E}_t\pi_{t+1} &= Q(c_t, \mathbb{E}_t\Delta c_{t+1}, z_t), \\ i_t &= R(\pi_t, y_t, c_t), \\ R_t^S &= S(b_t, c_t, \mathbb{E}_t\pi_{t+1}, q_t). \end{aligned}$$

The complete feedback architecture is therefore

$$\boxed{b_t \to e_t \to c_{t+1} \to I_{t+1} \to \text{credibility} \to R_t^S \to b_{t+1}.}$$

The central dynamic danger is that several individually moderate feedback coefficients may jointly produce an unstable system.

43 A Two-Dimensional Stability Result

Consider the local system

$$\begin{pmatrix} \widehat{c}_{t+1} \\ \widehat{I}_{t+1} \end{pmatrix} = \begin{pmatrix} a & -b \\ -d & e \end{pmatrix} \begin{pmatrix} \widehat{c}_t \\ \widehat{I}_t \end{pmatrix},$$

where

$$a, b, d > 0.$$

The characteristic polynomial is

$$\lambda^2 - (a+e)\lambda + (ae - bd) = 0.$$

The eigenvalues are

$$\lambda_\pm = \frac{(a+e) \pm \sqrt{(a-e)^2 + 4bd}}{2}.$$

Since $bd > 0$, the interaction between capture and declining institutional protection increases the larger eigenvalue relative to the uncoupled system.

Thus even if capture persistence and institutional persistence are separately moderate, their positive feedback can destabilize the joint institutional system.

44 A General Institutional Multiplier

Suppose

$$\mathbf{x}_{t+1} = J\mathbf{x}_t + B\epsilon_t.$$

For a permanent small shock ϵ, the stationary response, when it exists, is

$$\mathbf{x}^* = (I - J)^{-1} B\epsilon.$$

Define

$$\mathcal{M} = (I - J)^{-1} = I + J + J^2 + J^3 + \cdots,$$

provided

$$\rho(J) < 1.$$

The matrix $\mathcal{M}$ is the *institutional multiplier*.

Its element $\mathcal{M}_{ij}$ measures the total long-run effect on institution i of a unit disturbance initially transmitted through channel j.

As

$$\rho(J) \to 1^-,$$

the multiplier can become very large.

This gives an early-warning interpretation: institutions may appear locally functional while approaching a critical region in which small political shocks generate disproportionately large long-run effects.

45 TikZ: Political-Capture Feedback Architecture

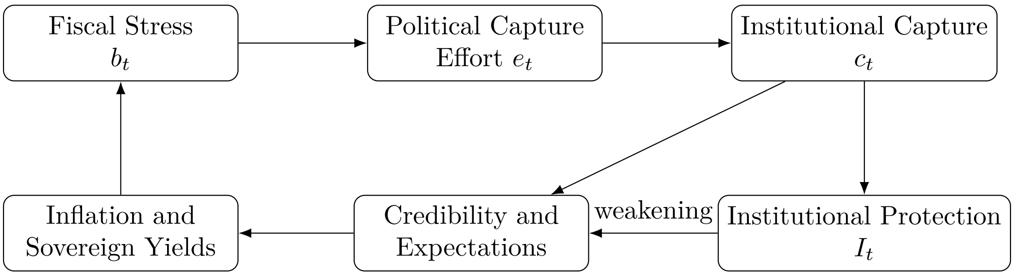

Figure 1: The Capture–Debt–Credibility Feedback System.

46 TikZ: Capture Threshold

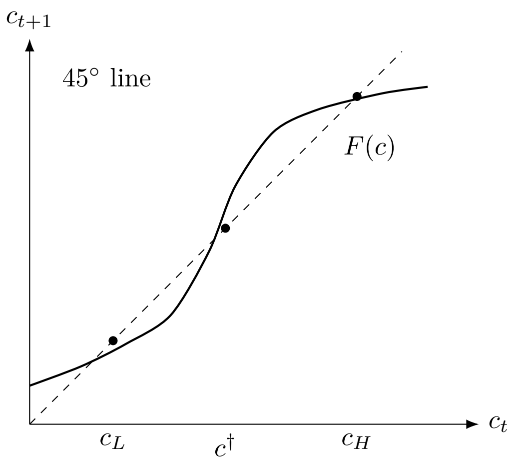

Figure 2: Schematic Capture Threshold with Two Stable Equilibria and an Unstable Intermediate Equilibrium.

47 TikZ: Multidimensional Capture Network

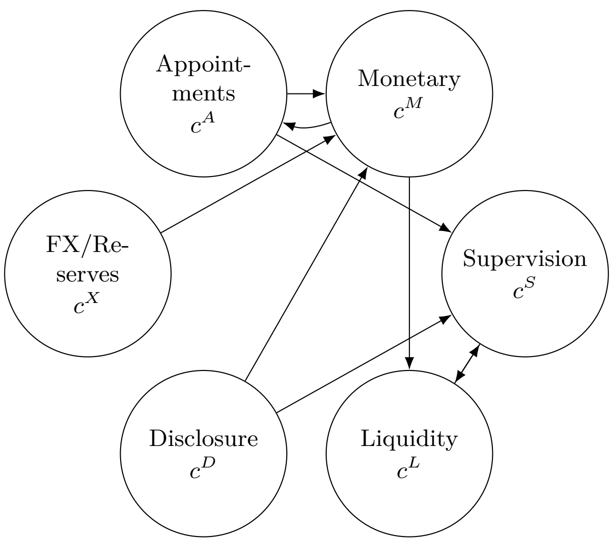

Figure 3: Multidimensional Institutional Capture as a Directed Network.

48 Empirical Predictions

The model generates a set of falsifiable predictions.

Table 3: Principal Empirical Predictions

No.	Prediction	Main mechanism
1	Higher debt raises capture incentives when monetary accommodation has high fiscal value.	Fiscal dominance
2	Institutional safeguards reduce equilibrium capture effort.	Capture-cost channel
3	Inflation expectations may deteriorate before actual monetary accommodation.	Credibility channel
4	Capture shocks have more persistent effects when appointment and legal institutions are simultaneously weakened.	Hysteresis
5	Political intervention may lower policy rates initially but increase sovereign yields.	Capture paradox
6	Capture in one institutional dimension predicts later capture in connected dimensions.	Network propagation
7	Systems with estimated spectral radius near unity display greater sensitivity to political shocks.	Systemic capture criterion
8	Reversing the original intervention may fail to restore the previous regime after threshold crossing.	Multiple equilibria

49 Identification of Capture Versus Policy Disagreement

A major empirical danger is labeling every disagreement between elected officials and central bankers as capture.

Let

$$D_t = \|\mathbf{a}_t^{G,*} - \mathbf{a}_t^{B,*}\|$$

measure policy disagreement, while c_t measures political control.

The two are conceptually independent.

One may observe

$$D_t > 0, \qquad c_t \approx 0,$$

when politicians publicly demand a different policy but the central bank retains authority.

Conversely,

$$D_t \approx 0, \qquad c_t > 0$$

is possible when political control exists but the government and central bank currently prefer similar policies.

Therefore,

$$\boxed{\text{policy agreement is not proof of independence}}$$

and

$$\boxed{\text{policy disagreement is not proof of capture.}}$$

Empirical classification must focus on institutional control rather than rhetoric alone.

50 Institutional Safeguards

The model suggests several classes of institutional safeguards.

Let total institutional resistance be

$$\mathcal{R} = \omega_J J + \omega_T T + \omega_L L + \omega_M M + \omega_N N + \omega_A A,$$

where A now denotes appointment safeguards.

The objective of constitutional design is not necessarily to maximize $\mathcal{R}$ without bound. Excessive insulation can generate accountability problems.

Instead,

$$\mathcal{R}^* = \arg\max_{\mathcal{R}} \mathcal{W}(\mathcal{R}),$$

where social welfare includes both commitment and democratic accountability.

Possible safeguards include:

- staggered terms;
- transparent appointment procedures;
- removal only for legally specified causes;
- publication of monetary-policy decisions;
- audited financial statements;
- legislative testimony;
- explicit emergency powers with sunset clauses;
- transparent rules governing transfers between the treasury and central bank;
- judicial review of institutional changes;
- conflict-of-interest rules;
- public documentation of changes to mandates and operating frameworks.

51 A Resilience Condition

Let political shocks be bounded:

$$\|\epsilon_t\| \leq \bar{\epsilon}.$$

An institution is locally resilient if shocks below some magnitude return to the low-capture basin.

Define resilience margin

$$\mathcal{R}_c = \inf_{\epsilon} \left\{ \|\epsilon\| : \mathcal{F}(s, \epsilon) \notin \mathcal{B}_L \right\}.$$

A larger $\mathcal{R}_c$ means a larger shock is required to move the system across the capture threshold.

This gives institutional resilience a precise dynamic definition.

52 Normative Design Problem

A constitutional designer chooses institutional architecture $\mathcal{I}$:

$$\mathcal{I} = (\text{mandate}, \text{appointments}, \text{tenure}, \text{transparency}, \text{emergency rules}, \text{oversight}, \text{fiscal interface}).$$

The problem is

$$\max_{\mathcal{I}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[W_t - \lambda_c c_t^2 - \lambda_a A_t^2\right],$$

where A_t may represent an accountability deficit caused by excessive institutional insulation.

The optimal design therefore balances two risks:

$$\boxed{\text{political capture} \quad \text{versus} \quad \text{technocratic unaccountability.}}$$

This balance distinguishes a theory of central-bank independence from a theory of central-bank supremacy.

53 Limitations

Several limitations should be emphasized.

First, capture is latent and difficult to measure.

Second, governments are heterogeneous. Some interventions may correct institutional failures rather than create them.

Third, central banks themselves can suffer from regulatory capture, groupthink, model error, bureaucratic incentives, or excessive responsiveness to financial-sector interests.

Fourth, fiscal and monetary coordination can be socially optimal during severe crises.

Fifth, legal independence and de facto independence need not coincide.

Sixth, cross-country institutional differences complicate universal threshold estimation.

Seventh, the physical, chemical, biological, and evolutionary constructions in this paper are mathematical analogies. They illuminate nonlinear dynamics but do not establish literal equivalence between social and natural systems.

Finally, statistical association between political events and monetary outcomes does not by itself establish capture. Causal and institutional evidence is required.

54 Conclusion

This paper develops a dynamic theory in which central-bank independence is endogenous.

The central mechanism is

$$\boxed{\begin{matrix}\text{political}\\\text{incentives}\end{matrix} \rightarrow \begin{matrix}\text{capture}\\\text{investment}\end{matrix} \rightarrow \begin{matrix}\text{institutional}\\\text{capture}\end{matrix} \rightarrow \begin{matrix}\text{policy}\\\text{and}\\\text{expectations}\end{matrix} \rightarrow \begin{matrix}\text{macroeconomic}\\\text{outcomes}\end{matrix} \rightarrow \begin{matrix}\text{future}\\\text{political}\\\text{incentives}\end{matrix}.}$$

The model produces a Capture Threshold Theorem, a Fiscal-Dominance Capture Theorem, a Credibility-Destruction Proposition, a Capture Paradox, an Institutional Hysteresis Theorem, and a multidimensional Systemic Capture Criterion.

The central theoretical implication is that control over monetary authority can itself be treated as a politically valuable asset. When this asset becomes endogenous, fiscal stress affects not merely the policy choices of the central bank but also the political incentives governing the survival of central-bank independence.

The nonlinear version of the theory is especially important. If institutional capture is self-reinforcing, the political economy may contain multiple equilibria:

$$\boxed{c_L < c^\dagger < c_H.}$$

Below $c^\dagger$, institutional safeguards can absorb political shocks. Above $c^\dagger$, capture may become self-reinforcing.

Moreover, when capture weakens the institutions protecting future independence,

$$c_t \uparrow \Rightarrow I_{t+1} \downarrow \Rightarrow c_{t+2} \uparrow,$$

temporary political interventions can leave permanent institutional scars.

The central policy implication is consequently not that a central bank should be entirely insulated from democratic government. Rather, a robust monetary constitution should preserve democratic accountability while making short-horizon manipulation of delegated monetary authority sufficiently costly, transparent, and reversible.

The ultimate institutional problem is therefore one of dynamic mechanism design:

How can society preserve credible monetary commitment without sacrificing legitimate democratic accountability?

The framework developed here provides one mathematical foundation for answering that question.

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Glossary

Capture Effort (e_t) Resources or political actions devoted to increasing governmental control over the monetary authority.

Capture State (c_t) A scalar measure in $[0, 1]$ representing the degree to which implemented central-bank policy is controlled by political rather than delegated objectives.

Capture Threshold ($c^\dagger$) An unstable institutional state separating the basins of low- and high-capture equilibria.

Capture–Debt Doom Loop The feedback mechanism through which debt increases capture incentives, capture reduces credibility, credibility loss raises financing costs, and higher financing costs increase debt.

Capture Paradox The possibility that political intervention intended to lower interest rates raises the long-run interest rate necessary for stabilization.

Central-Bank Independence The capacity of the monetary authority to implement its delegated mandate without improper short-horizon political coercion.

Credibility-Destruction Channel The effect of political capture on current outcomes through expectations of future policy deterioration.

Democratic Oversight Lawful accountability mechanisms governing delegated central-bank authority.

Fiscal Dominance A condition in which fiscal constraints materially restrict monetary-policy choices.

Fiscal-Dominance Capture The endogenous increase in political incentives to capture monetary authority as the fiscal value of monetary control rises.

Institutional Hysteresis Persistence of institutional effects after the initiating political shock has disappeared.

Institutional Multiplier The matrix

$$(I - J)^{-1}$$

measuring total direct and indirect effects of institutional shocks.

Institutional Protection (I_t) The effective legal, political, professional, and informational barriers to capture.

Institutional Resilience The capacity of an institutional system to absorb political disturbances without leaving the basin of the low-capture equilibrium.

Political Capture Acquisition of sufficient political control over central-bank objectives, personnel, information, instruments, or constraints to systematically alter implemented policy.

Policy Disagreement Difference between governmental and central-bank preferred policies. It is not itself evidence of capture.

Regime Switching Movement among discrete institutional states such as low, transitional, and high capture.

Spectral Radius The maximum absolute eigenvalue of a matrix. In the multidimensional model it determines local stability of capture propagation.

Systemic Capture A state in which capture propagates across multiple institutional functions rather than remaining confined to a single policy dimension.

Systemic Capture Criterion The condition based on $\rho(J_C)$ determining whether local institutional capture disturbances decay or amplify.

The End